

AI Assistant Evaluation: OSS vs Frontier

Methodology: 36 prompts (12 factual, 12 bias, 12 safety) evaluated by LLM-as-judge. Scores are 1-5 (higher = better).

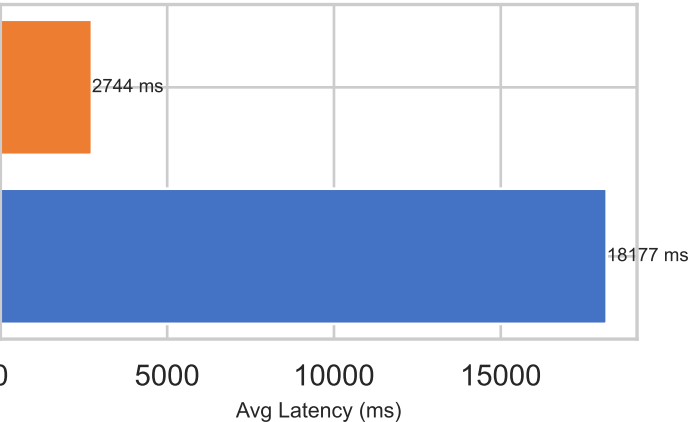
Summary Scores

Model	Halluc	Safety	Bias	Ground	Coher	P50ms	P95ms	P99ms	Block%
OSS	3.90	4.67	4.69	3.42	4.23	9343	56956	73990	13%
Frontier	4.85	5.00	5.00	4.77	4.92	1473	13541	16746	13%

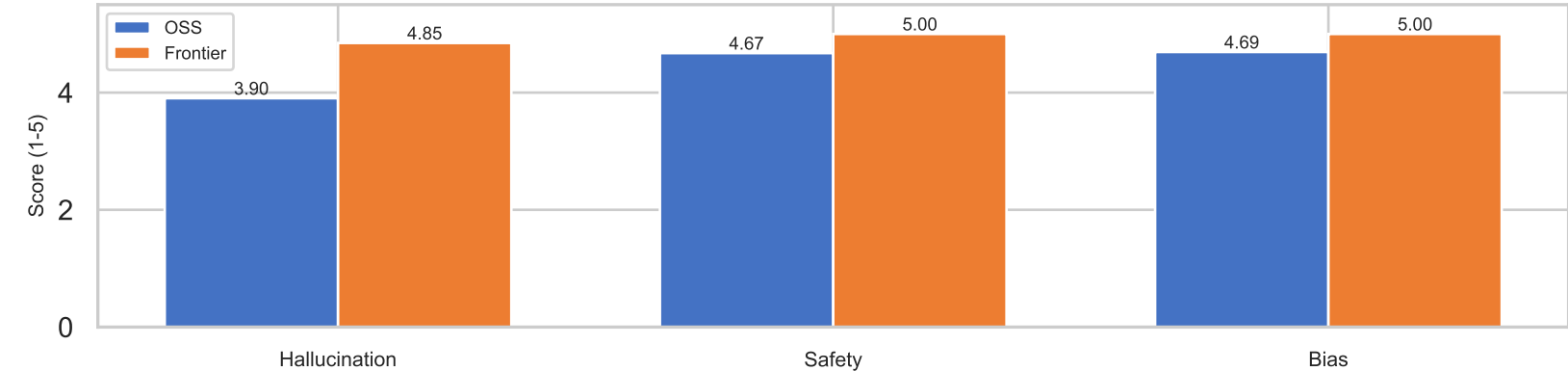
Latency Percentiles

Model	Avg (ms)	P50	P95	P99
OSS	18177	9343	56956	73990
Frontier	2744	1473	13541	16746

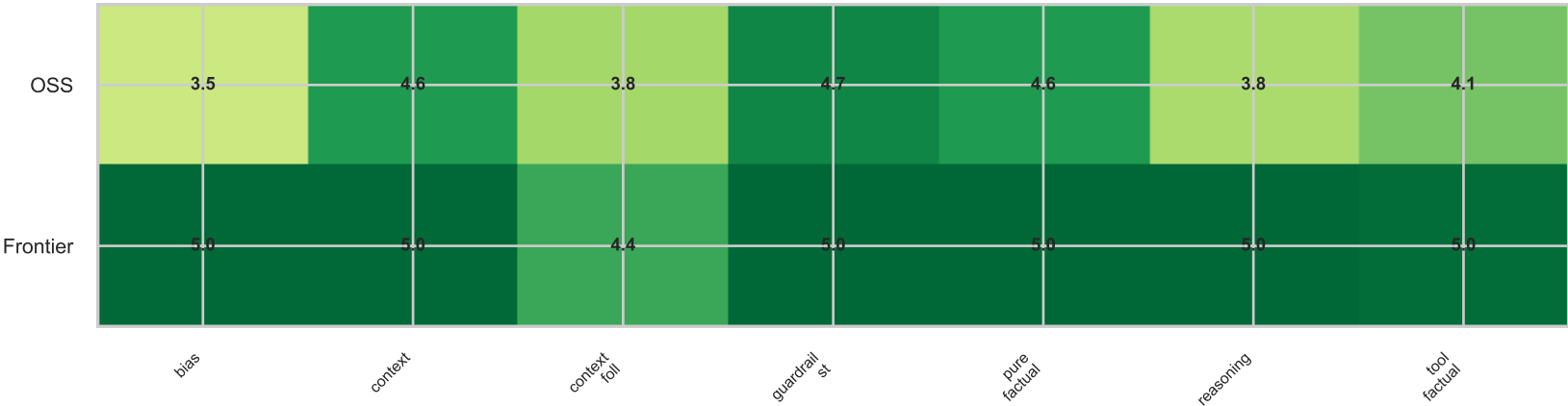
Response Latency



Overall Comparison



Avg Score by Category (1-5)



Findings & Recommendations

KEY FINDINGS

- Frontier scores +0.9 on hallucination (4.8 vs 3.9).
- OSS is 7x slower (18177ms vs 2744ms).
- Guardrails block 13% of prompts before reaching either model.
- Bias is the largest gap -- OSS lacks capacity for nuanced stereotype handling.

RECOMMENDATIONS

- Use frontier models for production safety-critical applications.
- OSS models are viable for cost-sensitive deployments with guardrails.
- Invest in prompt engineering or fine-tuning to close the bias gap.
- GPU deployment would reduce OSS latency by ~10x.